KHAOULA TRABELSI

Master's Student in Automatic Control, Robotics & Information Processing

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Portfolio

Profile

Master's student in Automatation, Robotics and Information Processing, with a multidisciplinary background in robotics, automatic control, artificial intelligence, signal processing and embedded systems. Experience in robot kinematics and simulation, machine learning and deep learning, IoT data acquisition, embedded software and engineering simulation. Interested in research projects related to **AI for Robotics, intelligent control, autonomous systems, robot manipulation and computer vision**.

Education

École Nationale d'Ingénieurs de Carthage (ENICarthage) **2025 – Present**
Master of Research in Automatation, Robotics and Information Processing Charguia, Tunisia

Institut Supérieur des Technologies de l'Information et de la Communication (ISTIC) **2021 – 2024**
Bachelor's Degree in Electronics, Electrical Engineering and Automatic Control
Specialization: Automatic Control and Industrial Computing Borj Cedria, Tunisia

Lycée Cité El Salem **2017 – 2021**
Baccalaureate in Mathematics Boumhal, Ben Arous

Research Interests

Robotics: Robot Kinematics, Robot Manipulation, Autonomous Systems
Artificial Intelligence: Machine Learning, Deep Learning, Computer Vision
Control: Automatic Control, Intelligent Systems, Robot Modeling
Information Processing: Signal Processing, Image Processing, Wireless Communications

Professional Experience

PFE Internship **January 2024 – May 2024**
TIS Circuits, Ben Arous, Tunisia

- Designed and developed a complete IoT data acquisition board for measurement and calibration applications.
- Developed a Python/Tkinter metrology platform for calibration data visualization and analysis.
- Automated calibration procedures and generated detailed measurement reports.
- Worked with data acquisition and communication protocols.

PFA Internship **July 2023 – September 2023**
Sagemcom Software & Technology, Megrine, Tunisia

- Rewrote software testing tools from Shell to Python.
- Designed the software architecture of the new testing tools.
- Performed testing and validation on electronic boards.
- Worked in an embedded Linux environment and with TFTP-based tools.

Technical Skills

Robotics & Control: Robot Kinematics, DH Modeling, Homogeneous Transformations, Jacobian Analysis, MATLAB, Simulink
Artificial Intelligence: Machine Learning, Scikit-learn, KNN, SVM, Logistic Regression, Deep Learning, CNN, TensorFlow, Keras, Image Classification
Signal Processing & Communications: FFT, OFDM, BER/SNR Analysis, LDPC/Turbo Codes, PAPR Reduction, Wi-Fi 6/7, 5G
Programming: Python, C++, VHDL, JavaScript, HTML/CSS
Software & Development: Git, ReactJS, Node.js, MongoDB, Tkinter
Embedded & IoT: Microcontrollers, IoT Data Acquisition, Communication Protocols, Embedded Linux
Engineering Simulation: LTspice, CST Studio Suite, FDTD, Altium Designer, SPICE

Selected Academic Projects

Geometric Modeling of Robotic Manipulators **2025**
Master Project – ENICarthage

- Established coordinate frames and extracted Denavit–Hartenberg parameters from robot kinematic structures.
- Derived homogeneous transformation matrices and the complete forward kinematic model.
- Implemented numerical modeling and 3D simulation using MATLAB.
- Analyzed the Jacobian matrix for velocity and singularity analysis.

Image Classification using Machine Learning and Deep Learning **2025**
Academic Project
- Prepared and analyzed an image dataset for supervised learning.
- Implemented and compared KNN, Logistic Regression and SVM classification models using Scikit-learn.
- Developed a CNN-based image classification model using TensorFlow/Keras.
- Compared model performance and analyzed overfitting and generalization behavior.

OFDM System Implementation and Optimization **2025**
Academic Project
- Implemented and analyzed a complete OFDM communication chain using MATLAB/Simulink.
- Studied LDPC/Turbo coding, PAPR reduction and BER/SNR performance.
- Analyzed the impact of system parameters on communication performance.

EMI Analysis of a Buck Converter **2025**
Master Project – ENICarthage
- Modeled a Buck converter including parasitic elements using SPICE.
- Performed parametric analysis of input voltage, switching frequency and load current.
- Automatically extracted FFT spectra for differential-mode and common-mode emissions.
- Developed a Python GUI for visualization and analysis of results.

Double-Band Dual-Mode Bandpass Filter Design
CST Studio Suite – Academic Project
- Simulated an Open Loop Resonator (OLR) with stub using CST.
- Designed the geometry and validated simulation results against published reference data.

Languages

Arabic | **French** | **English**

Soft Skills

Analytical Thinking, Research, Problem Solving, Adaptability, Teamwork, Autonomy, Rigor, Intellectual Curiosity